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27 August 1958
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3 July 1951

MILITARY SPECIFICATION

COMPRESSOR UNIT, AIR, AIRCRAFT, SHAFT POWER DRIVEN,
GENERAL SPECIFICATION FOR

1. SCOPE

1.1 This specification covers the requirements for all types of shaft power driven aircraft air compressor units that are used to supply high pressure air for pneumatic systems conforming to the requirements of Specification MIL-P-5518.

1.2 CLASSIFICATION.- The type designation will be assigned by the Government and shall be specified in the model specification.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATION

Military

MIL-P-116	Reservation, Methods of
MIL-D-5028	Drawings and Data Lists; Prepartion of' Manufacturers$ (For Production Aeronautical end Associated Equipment)
MIL-E-5272	Envimmnental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-F-7105	Ripe Threads, Taper, Aeronautical National Form, Symbol ANFT
MIL-7742	Screw Threads; Standard, Aeronautical
MIL-F-936	Parts and Equipment, Aeronautical Pre-paration for Delivery

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MS33540	Safety Wiring - General Practices For

Air Force - Navy Aeronautical

AN995	Wire - lock

FSC 1650

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AN4047	Gasket - Type XII 01' Type XIV A and B Engine Accessory Drive
AND10050	Bosses - Standard Dimensions for Gesket Seal straigh Thread
AND10262	Flange - Type XII Accessory Mounting

PUBLICATIONS

Air Force-Navy Aeronautical

No. 143	Specifications and Standards, Use of

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring actitity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 PREFRODUCTION TESTS.- This specification makes provisions for preproduction testing.

3.2 MODEL SPECIFICATION.-A shaft power driven air ccmpressor unit model specification in accordance with appendix 1, shall be submitted by the contractor for approval prior to initiation of preproduction test.

3,3 SELECTION OF SPECIFICATIONS AND STANDARDS.- Specifications and standards for necessery commodities and services not specified herein shall be selected in accordance with ANA Bulletin 143 except as providedin 3.3.1 and 3.3.2.

3.3. COMMERCIAL PARTS. - Commercial parts having suitable proporties may be used where on the date of initation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.3.2 STANDARD PARTS.- With the exception in 3.3.1, or AN Standard parts shall be used where they suit the propose. They shall be identified on the drawings by their part numbers.

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3.4 MATERIALS.- Material and processes used by the manfacturer of' compressor units shall be of high quality, suitable for the purpose, and shall conform to Government specifications. The contractors specifications may be used, provided the specifications are released by the Services and contain provisions for adequate tests. The use of the contractor's specifications till not constitute waiver of acceptance inspection.

3.5 DESIGN AND CONSTRUCTION

3.5.3 DIMENSIONs. - The configration, dimensions, size and location of ports of the compressor unit shall be specified in the model specification.

3.5.2 GASKEKS.- Each compressor unit shall be provided with a mounting flange gasket as specified on Standard AN4047.

3.5.3 MOUNTING POSITION.- The mounting position shall be specified in the model specification.

3.5.4 MOUNTING FLANGE. - The mounting flange shall conform to Standerd AND10262. The flange shall be made from material that will resist fatigue from vibration and reverse stresses set up by torsional vibration from the engine driven shaft.

3.5.5 SAFETYING.- All threaded parts shall be securely locked or safetied by safety-tiring, self-locking nuts or other approved methods. Safety wire shall conform to Standard AN995 and shall have a minimum diameter of 0.032 inch. The wire shall be applied in accordance with the practice outlined in Standard MS33540.

3.5.6 DRAIN BOSSES.- An external boss in accordance with Standard AND10050 shall be included on the compressor crankcase for returning oil to the engine for propulsion engine compressors. When oil is supplied by other external sources, inlet and outlet port bosses shall be in accordance with Standard AND10050.

3.5.7 LUBRICATION LEAKAGE. - There shall be no lubrication fluid drippings from the unit. In the event that a drainage means iS required, a drain boss in accordance with Standard AND10050 for a 1/4 inch tubing shall be provided for attachment of a drain lines

3.5.8 BALANCE. - The moving partss of the compressor unit shall be balanced and the compressor shall not cause destructive vibrations under any conditions of operation.

3.6 PERFORMANCE.- The compressor unit shall be capable of operating satisfactorily under the following conditions:

3.6.1 OPERATING Range.- The unit shall be designed to operate satisfactorily throuout the range specified in the model specification.

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3.6.2 CAPACITY.- The rated capacity and the minimum capcity of the unit shall be specified in the model specification at rated conditions and at varicus conditions of inlet and outlet pressures, speed, ambient pressures and temperatures.

3.6.3 OIL CONSUMPTION.- The oil consumption shall not exceed 20 cc per hour regardless of the lubrication method used.

3.6.4 TORQUE.- The torque required to drive the compressor unit shall not exceed that specified in the model specification.

3.6.5 RATED CONDITION

3.6.5.1 RATED SPEND.- The rated speed shall be specified in the model specification.

3.6,5.2 RATED PRESSURES.- The rated inlet and delivery pressure shall be specified in the model specification.

3.6.5.3 RATED LIFE.- The rated life at rated conditions shall be specified in the model specification.

3.6.6 DUYT CYCLE.- The duty cycle shall be specified in the model specification.

3.6.7 LUBRICATIION.- The copressor unit shall operate satisfactorily when oil is supplied by the prime mover through the mounting flange. The required oil flow and pressure shall be specified in the model specification. The type of oil shall be specified in the model specification. (The above is not applicabe to compressor units with self-contained lubrication systems.)

3.6.8 COOLING,- The compressor unit cooling requirements shall be specified in the model specification.

3.6.9 INVERTED FLIGHT, - The unit shall be capavle of operation in an inverted position fcr a period of 60 seconds.

3.6.10 ENVIRONMENTAL.- The unit shall be capable of satisfactory operation during and after exposure to any of the following conditions:

a. Temperature ranging from -65°F ± 5°F, without benefit of solar radiation, up to the maximum temperature as specified in the model specification. However, this maximum temperature shall not be less than 160°F plus or minus 5°F.

b, Fungus growth as encountered in tropical climates.

c. Exposure to salt-sea atmosphere.

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d. Relative humidity of 95 ± 5 percent at a temperature of 160° ± 5°F.

e. Exposure to airborne sand and dust particles encountered in normal and desert operation.

f. Vibration incident to service use.

g. Mechanical Shock.- The magnitude and duration of the shock, the number of shocks and position of the test specimen shall be specified in the model specification.

h. Thermal Shock.- The requirements for thermal Shock shall be specified in the model specification.

3.7 INTERCHANEABILITY.- All parts having the same manufacturer's part number shall be functionally and dimensionally intechangeable. (Changes in the manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.8 WEIGHT.- The weight shell be kept to a minimum consistent with the requirements of this specification. The dry weight of the assembled compressor unit shall be specified in the model specification.

3,9 THREADA.- Only straight threads conforming to Specifications MIL-S-7742 shall be used. If desired, pipe threads conforming to Specification MIL-P-7105 may be used for connections within the unit that are not broken for the life of the unit, or for pemanently plugged, drilled, or cored openings.

3.10 IDENTIFICATION OF PRODUCT

3.10.1 MARKING OF PARTS.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.10.2 NAMEPLATE.- A nameplate, permanently and legibly marked with the information listed below, shell be securely attached to each compressor unit. The information marked in the spaces provided on the nameplete shall be in accordance with MIL-D-130.

Compressor Unit, Air, Aircraft, Shaft Fewer Driven
Type (insert Service type designation)
Rated Pressure, Air, psi
Rated Capacity, Air, lb/min
Rated Speed, _ rpm
Model Specification No.
Manufacturer's Part No.
Manufacturer's Serial
Order No. ____
Manufacturer's Name or Trademark
U. S.

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3.11 WORKMANSHIP.- The workmanship and finish shall be of sufficiently high grade to insure satisfactory reliability and durability consistent with the servicelife and application of the compressor unit.

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS.- The inspection and testing of shaft power driven compressor units shall be classified as follows;

a. Acceptance tests	4.2
b. Preproduction testing	4.5

4.2 ACCEPTANCE TESTS.- The acceptance tests shall consist of:

a. Individual test

b. Periodic Sampling Plans and Tests

4.2.1 INDIVIDUAL TESTS.-. Each compressor unit submitted for acceptance under contract shall be subjected to the following tests as described under 4.4 TEST METHODS, and shall be conducted in the order listed unless otherwise specified:

Examination of product	4.4.1
Break-in run	4.4.2
Capacity	4.4.3
Oil consumption	4.4.4
Torque Measurement	4.4.5

4.2.2 PERIODIC SAMPLING PLANS AND TESTS.- The following number of compressor units shall be selected.

a. One comprosser unit shall be selected at random from the first 50 units, or frection thereof, produced on contract.

b. One compressor unit selected at randcm from each 200 units, or fraction thereof during the production period.

The selected compressor units shall be subjected to the following testes

Endurance	4.4.10
Recalibration	4.4.11

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4.2.2.1 REJECTION AND RETEST.- When one item selected from a production run fails to meet the specifications no items still on hand or later produced shall be accepted until the extent end cause of failure are determined.

4.2.2.2 INDIVIDUAL TESTS MAY CONTINUE.- For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or produced later shall not be made until it is determined that items meet all the requirements of' the specification.

4.2.3. DEFECTS IN ITEMS ALREdY ACCEPTED.- The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and methods of correcting them.

4.3 TEST CONDITIONS

4.3.l MEASUREMENTS. AND APPARATUS.- The instrumentation used for pressures, temperetures, speed, torque, and other parameters specified in the model specification, shall make possible easy readings with an accuracy of $\pm$ 2 percent.

4.3.2 LABORATORY TECHNIQUE.- The manufacturer shall submit detail plans of his test setup to the procuring activity before he conducts the preproduction tests. Precise laboratory procedures shall be adhered to. Temperatures shall be picked up at a location in the center of the air stream where the influence of variations of accidentally variable outside factors are least felt. The outlet temperatures shall be picked up in the center of the stream as closely as possible to the outlet valve, and the thermocouple wires shall be as thin as practical and as protected from outside variable cooling factors as possible in order to obtain accurate readings.

4.3.3 FILTRATION.- For all tests, a 10-micron filter shall be installed at the inlet of the air compressor unless a filter is furnished as an integral part of the compressor

4.3.4 AMBIENT CONDITIONS.- Unless otherwise specified, the ambient pressure shall be 27 inches to 30 inches of mercury absolute and the ambient temperature shall be 70°F to 100°F

4.4 TEST METHODS

4.4.1 EXAMINATION OF PRODUCT.- Each compressor unit shall be carefully examined to determine conformance with requirements or this specification with respect to dedign. weight, dimensions, uorkmanship. marking, and for my visible defects.

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4.4.2 BREAK-IN RUN. - The break-in run shall be conducted as specified by the contractor in the model specification.

4.4.3 CAPACITY.- At rated compressor speed, the capacity of the compressor shall be measured and shall fall within the required limits as specified in the model specification.

4.4.4 OIL CONSUMPTION.- Each compressor shall be run at rated speed with rated air inlet pressure, inlet temperature at 100 degrees plus or minus 5 degrees Fahrenheit, and air delivery pressure as specified in the model specification. For externally lubricated compressors the inlet oil pressure mall be 2.5 ± 0.5 psi. Each compressor shall be operated for 1 hour to determine the amount of oil consumed. The shall be done by using a suitable metering system, taking care that all oil vapors are trapped in the vent line and returned to the metering tank. The pressure of the oil entering the compressor shall then be increased to 25 plus or minus 0.5 psi and the test repeated for eternally lubricated compressors* Oil consumed for each run, in cubic cenitimeters per hour, shall be recorded. The oil consumption shall also be determined with crank-case pressure at 5.5 inches of mercury absolute.

4.4.5 TORGUE MEASUREMENTS - The torque required to drive the compressor under conditions specified in the model specification shall be measured.

4.4.6 CAPACITY RUNS.- The capacity at ambient temperature shall be determined during the following:

a. Runs shall be made at 25, 5O, 75, 100 and 110 percent of rated speed, at zero, 25, 50, 75, and 100 percent rated air delivery pressure while maintaining rated air inlet pressure,

b. At rated air discharge pressure and rated speed the air inlet pressure shall be varied throughout the operating range as specified in the model specification. The capacity of at least four air inlet pressure operating points, including tine minimum and maximum points, shall be measured during this run.

4.4.7 PROOF PRESSURE.- The unit shall be run at rated speed, and at all other rated conditions as specified in the model specification. The discharge pressure shall be allowed to build up to 150 percent of rated delivery pressure with the reservoir-tubing volume sized so that time for building from 100 percent oc 150 percent of pressure shall not exceed 1 minute. The unit shall then be run at 150 percent delivery pressure for an additional period of 1 minute. If desired, the 150 percent rated outlet pressure may be obtained with an external source of pressure with the unit in a static condition.

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4.4.8 IOW TEMPERATURE STARTING.- The unit shall be placed in a cold box and bolted to a drive pad. A thermocouple shall be attached at a suitable point on the compreassor. A reservoir shall be used at the delivery side of the air compressor. The reservoir capacity shall simulate the conditions for the intended compressor installation. The compressor shall be cooled to minus 65°F plus or minus 5° and soaked for 12 hours after temperature stabiltzation. The compressor reservoir pressure shall not be less than 80 percent of rated delivery pressure. The compressor shall be started and allowed to pump the reservoir to rated pressure after which it shall continue to run at rated delivery pressure. Total running time from start to finish shall be 15 minutes. The unit shall be stopped and again allowed to cool to minus 65° plus or minus 5°F with a l-hour soak after temperature stabilization. The starting procedure shall be repeated. A total of 10 starts, with 1-hour stabilized soak for each start after the first start shall be made.

4.4.9 TEMPETATURE.- The unit shall be soaked for 12 hours at the maximum temperature at which the unit is capable of operating as specified in the model specification. The unit shall be operated at that temperature and see level pressure at rated delivery pressure and rated speed for a period representative of the cOntinous operating capability of the unit at these conditions.

4.4.10 ENDURUNCE.- The endurance test on the unit shall be conducted in accordance with table I. The air volume of the test set up including air lines, surge tanks, etc, shall simulate the intended compressor installation.

TABLE I

Percent of Rated Life (Hours)	Percent of Rated Speed	Inlet Pressure
10	90	Minimum value as specified in the model specification
10	110	Maximum value as specified in the model specification
80	100	Rated

The duty cycle shall be representative of the Intended compressor application.

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4.4.10.1 WEAR ANALYSIS.- At the end of endurance test, the unit shall be thoroughly inspected. Excessive deteriration or breakage of parts shall be cause for rejection. If, during the endurance testing a major part fails. the test shall be terminated and a replacement unit shall be submitted for a complete endurance test. The replacement unit shall have a redesigned part or one of different material corresponding to the failed part except that, if the failure was caused by faulty material or workmanship the procuring actitity my authorize the installation of a part of the original design and material with the defect overcome. The endurance test shall be considered completed when all major parts have concurrently completed the endurance test. Minor parts may be replaced and penalty runs conducted at the option of the procuring actitity.

4.4.11 RECALIBATION.- Following the endurance test, the compressor shall be recalibrated and the capacity shall be as specified in the model specification. The compressor capacity shall not have decreased more than l5 percent from the capacity prior to the endurance test at the condition defined.

4.4.12 ENVIRONOMENTAL

4.4.12.1 PREFORMANCE RECORD.- Prior to conducting any of the tests specified herein the equipment shall be operated under room conditions and a record made of all date necessary to determine compliance with the contractor's model specification. These data shall provide the criitetia for checking satisfactory performance of the equipment undergoing enviomental tests. Variation from room condition performance shall be within limits acceptable to the procuring activity. Where the contractor's model specification establishes the level of acceptable performance for the test procedure, a comparsion with room condition perfomance shall not be made.

4.4.12.2 DETERIORATION.- Deterioration or corrosion of any intermel or external components which could in any manner prevent the equipment from meeting operational maintenance and servicing requirements during service life shall provide reason to consider the equipment as having failed to meet the test to which it was subjected.

4.4.12.3 TESTS

4.4.12.3.1 FUNGUS.- The compressor unit shall be tested in accodance with proceure I of Specification MIL-E-5272.

4.4.12.3.2 SALT SPRAY.- The unit shall be tested in accordane with procedure I of specification MIL-E-5272.

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4.4.12.3.3 HUMIDITY.- The unit shall be tested in accordance with procedure I of Specification MIL-E-5272.

4.4.12.3.4 SAND AND DUST.- The unit shall be tested in accordance with procedure I of Specification MIL-E-5272.

4.4.12.3.5 VIBRATION.- The unit shall be tested in accordance with procedure I of Specification MIL-E-5272 with the exception that the frequency range and gravity loading shall be specified in the model specification.

4.4.12.3.6 MECHANICAL SHOCK.- The unit shall be tested in accordance with procedure I of Specification MIL-E-5272.

4.4.12.3.7 THERMAL SHOCK,- The thermal shock tests shall be specified in the model Specifications

4.5 PRREPRODUCTION TESTING

4.5.1 PREPROUCTION TEST SAMPLE.- The preproduction test samples shall ccnsist of models representative of the production equipment. They shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.5.2 PREPRODUCTION TESTS.- The preproduction tests shall consist of all tests described under TEST METHODS oonducted in the order as listed.

4.5.3 DATA ACCOMPANY THE PREPRODUCTION SAMPLE

4.5.3.1 PREPRODUCTION TEST REPORT.- The test report submitted with the preproduction test sample shall include the following information:

a. Report of all tests, graphically pretendted when appropriate

b. Summery of all tests with conclusions and recommendations

c. Diagram of all test setups with a list of all manufacturers and part numbers of components used

d. Detail description of all tests and test conditions

e. Copies of test log sheets

f. Photographs When appropriate

The test report shall contain the results of the capacity test runs and shall be graphically presented with capacity, torque, volumetric efficiency. air inlet and delivery pressure as veriables in apporiate

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form. The report shall also contain a tabulation of speed, islet pressure, outlet pressure, temperature cf each compressor stage, bulk oil temperature, shaft seal leakage, and ambient temperature for each of the conditions listed in table I. The readings shall be recorded at every 12-hour period, but not less than twice for each test run.

4.5.3.1.1 The test report shall be submitted in duplicate arid shall contain original data sheets or photostatic copies thereof. All requests for preproduction approval shall be accompanied by one set of detail and assembly drawings.

4.5.3.2 DRAWINGS TO ACCOMPANY THE PRODUCTION SAMPLE. - The assembly drawings shall consist of a cutaway section showing all details in their normal assembly position and shall carry pert numbers of all details and subassemblies. The following data shall be furnished on or together with all assembly drawings:

a. Mounting dimensions

b. Port dimensions

c. Overall dimensions

5. PREPARATION FOR DELIVERY

5.1 PREERVATION, PACKAGING AND PACKING.- Air compressor units shall be prepared for delivery in accordance with Specification MIL-P-7936. Preservation for level A shall method II of Specification MIL-P-116.

5.2 MARKING FOR SHIPMENT AND STORAGE.- Interior packages and exterior shipping containers shall be marked in accord with MIL-STD-129, The nomenclature shall “be as follows: Compressor Unit, Air, Aircraft Shaft Power Driven.

6. NOTES

6.1 INTENDEDUSE.-The compressor units covered by this specification are intended for use on aircraft to provide compressed air for operation of lending gears, bomb-bay doors, gun chargers, and flight control systems.

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6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date of this specification

b, Number of preproduction samples and point of testing

c. Applicable levels of preservation and packaging and packing

NOTICE: When Government- drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in my way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporaticn, or conveying any rights or pemission to manufacture use or sell any patented invention that may in my way be related thereto.

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APPENDIX I

MODEL SPECIFICATION FOR

OUTLINE AND INSTRUCTIONS FOR PREPARATION

10. SCOPE

10.1 This appendix establishes the form to be used by manufacturers in the preparation of a shaft power driven air compressor unit model specification.

20. APPLICATION

20.1 A complete compressor unit model specification in accordance with the "Outline and Instructions for Peparation" as specified herein, shall be prepared for each specific compressor unit model. New compressor unit model designtions will be assigned by the Goverment. Amendments to an approved compresson unit model specification which has been released will be acceptable to the procuring activity. Each amendment shall be approved and shall include and supersede the previous amendment.

20.2 The headings and numbering of sections and paragraphs herein correspond to those of the basic specification for the specific data needed only in the model specification. Omissiol of reference in the model specification to a particular requirement of the basic specification shall be interpreted as compliance therewith. When departures are necessary from the requirements of this appendix and the basic specification, the details of such departures shall be stated as specific requirements bearing the same section and paragraph heading and numbering as in the basic specification.

20.3 Parenthetical sentences, phrases, and words are included herein for the guidance of the manufacturer for insertion of proper information and data related thereto, in connection with the preparation of the model specification. Parenthetical statements shall not be copied verbatim in the model specification.

20.4 The spectification number shall be the number assigned by the manufacturer. When revisions are made, they shall be designated by the use of a dash and a letter following the number, with a reviaion date therefor, which shall be shown on page 1 only. Only the specification number and the revision suffix letter, if applicable, shall be shown on subsequent pages.

20.5 For purposes of pemitting preliminary evaluation of a proposed compressor unit design, or for release of approved compressor unit performance characteristics in connection with a design competition, the compressor manfacturer may submit a preliminary model specification to serve until superseded by a complete coordinated model specification which will be required for a production contract,

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30. MODEL SPECIFICATION

30.1 The format and instructions for the preparation of the model specification follow:

(Spec. No.)_____
(Date)_____
(a) Revised (date)
(b) Revised (date)

MODEL SPFECIFICATION

COMPRESSOR UNIT, AIR, AIRCRAFT, SHAFT POWER DRIVEN
(Insert activity model designation if assigned)

(NAME OF CONTRACTOR)
(Contractor's Model Designation)

1. SCOPE

1.1 SCOPE- This specificaton covers the requirements for a _____(insert activity type designation) shaft power driven sir compressor unit.

1.2 CLASSIFICATION.- The (_____ inert type) compressor for installation in aircraft (insert designation(s) of applicable aircraft) will be used to furnish high pressure air for (enumerate applications for which unit may be used.)

2. APPLICABLE DOCUMENTS

2.1 The following documents cf the issue in effect on date of invitation for bids, shall form a part of this specification:

SPECIFICATION

MIL-C-6388	Compressor Unit, Air, Aircraft, Shaft Power Driven, General Specification For

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.5.1 DIMENSION - The overall dimensions of the compressor, the size and location of the ports, and location of components

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are as shown in figure _____. The compressor shall contain the following necessary parts:

Unloader . . . . . Yes _____ No _____

Cooling Means: Blast Air . . . . . Yes No _____
Fan. . . . . Yes No
Liquid . . . Yes . . . . Yes No

3.5.3 MOUNTING POSITION,- The mounting position shall be as shown in figure

3.6.1 OPERATING.- The unit shall operate satisfactorily throught the following range of conditions

Ambient temperature . . . . . to _____

Altitude or Compartment Density . . . . . to _____

Inlet Pressure . . . . . to _____

Inlet Temperature.. . . . . to. _____

Speed.. . . . . to _____

3.6.2 CAPACITY.- The rated capacity of the compressor shall be _____ lbs/min ± _____ lbs/min at rated conditions. A graph showing minimum compressor capacity (lbs/min) versus discharge air pressure (psig) versuspeed versus inlet ar pressure (in. Hg. abs.) and temperature ishown on figure _____. (Other curves may be included as required.)

3.6.4 TORQUE.- The maximum required torque to drive the compressor is _____.

3.6.5.1 RATED SPEED.- The rated speed shall be

3.6.5.2 RATED PRESSURES.- The rated air inlet pressure and delivery pressure shall be ____Hg. aba, and _____psig respectively.

3.6.5.3 RATED LIFE.- The rated life at rated conditions in hours (Various life ratings at other than rated conditions may also be specified.)

3.6.6 DUTY CYCLE.- The "duty cycle" shall be as follows: (Insert amount of loaded and unloaded time per hour.)

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3.6.7 LUBRICATION.- (For oil supplied from a propulsion engine pad) the oil pressure available shall vary between psig and Maximum oil flow available will not exceed _____ccs/min. Oil shall conform to Specification _____.

3.6.8 COOLING.- (If integral cooling is not supplied) For air cooling. the following ambient temperature-altitude condition shall be applied during testing of the unit.

TABLE_____

Cooling System (if provided) Pressure Drop	Altitude or Compartment Density	Cooling Air Temperature	Flow lbs/min	Max. BTUS per Minute Rejected (where applicable)

For liquid cooling, (specify type of liquid)_____ having a flow of gpm. shall be used for cooling purposes. Pressure drop through the cooling shroud shall not exceed psig and temperature rise of the cooling liquid shall not exceed_____ °F. Installation Drawing Figure shows the manner in whcich the unit is installed for cooling provision. (Indicate configuration, airflow direction, maximum allowable compartment temperature, and other information peculiar, to the installation.)

3.6.10.g MECHANICAL SHOCK.- The magnitude and duration of the mechanical shock shall be_____ number of shocks _____, and position of the test specimen _____.

3.6.10.b. THERMAL SHOCK.- The requirments for thermal shock shell be as follows: _____e

3.8 WEIGHT.- The maximum dry weight of the compressor unit shall not exceed pounds.

4. QUALITY ASSURANCE PROVISIONS

4.4.2 BEAK-IN RUN.- The break-in run shall be as follows _____. (Insert test procedure and compressor operating conditions.)

4.4.5 TORGUE MEASUREMENT.- (specify conditions)

4.4.12.3.5 VIBRAITON.- The frequency range and gravity loadins is

4.4.22.3.6 THERMAL SHOCK. - (Insert test procedure.)

5. PREPARATION FOR DELIVERY

5.2 (The contractor shall specify the packaging requirements for storage and shipment of the compressor unit for approval by the procuring activity.)

6. NOTES

6.1 INTENDED USE.- (The contractor shall give a brief description of the intended use of the compressor unit.)

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